

Real-Time Landslide Risk Prediction Using Soil Moisture and Micro-Vibration Sensors with Edge-Based Random Forest Classification

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Abstract—Landslides are among the most destructive natural hazards in mountainous and monsoon-prone regions, causing severe loss of life and infrastructure damage. This paper presents a low-cost real-time landslide risk prediction system using soil moisture and micro-vibration sensors integrated with an Arduino Nano. A dataset of 894 samples was collected under varying environmental conditions and analyzed using machine learning techniques to understand the nonlinear relationship between soil moisture and ground vibration. Based on the trained model insights, optimized threshold rules were derived for computationally efficient edge deployment. The final embedded system classifies slope conditions into stable, weakening, and critical states in real time without cloud dependency. Experimental results show that the offline machine learning model achieved 94.2% testing accuracy, while the embedded rule-based implementation achieved 92.8% accuracy under real-time hardware validation. The proposed framework demonstrates that machine learning-assisted threshold optimization enables reliable, low-power, and deployable early warning systems for landslide-prone regions.

Index Terms—Landslide detection, Soil moisture sensor, Vibration sensor, Random Forest, Edge computing, Arduino Nano, Real-time monitoring

I. INTRODUCTION

Landslides are one of the most destructive natural hazards, particularly in mountainous and monsoon-prone regions. They cause significant loss of life, infrastructure damage, and environmental degradation. Early detection and real-time monitoring of slope instability are essential to reduce disaster impact. Recent advancements in Internet of Things (IoT), wireless sensor networks, machine learning (ML), and remote sensing technologies have significantly improved landslide prediction and early warning systems.

IoT-based sensing platforms enable real-time environmental monitoring in disaster-prone regions. Edge-intelligent frameworks for ultra-low memory IoT nodes have been proposed to support adaptive and resource-efficient disaster monitoring systems [1]. Geophone-based microseismic precursor detection systems [2] and IoT-driven slope monitoring platforms deployed in landslide-prone regions [3] demonstrate the importance of real-time sensing. Data-driven disaster analytics frameworks further enhance decision-making capabilities [4].

GIS-based and machine learning-based susceptibility mapping approaches have improved spatial prediction accuracy [5]–[9]. Remote sensing and satellite-driven detection models have also been widely explored [10]–[16]. Vegetation indices,

rainfall data, and multi-modal earth observation techniques have been integrated for improved prediction [17]–[19]. Advanced ML algorithms such as Support Vector Machines, fuzzy logic models, transfer learning, and deep neural networks further enhance susceptibility assessment [20]–[29].

However, most existing approaches rely on cloud processing, high computational resources, or large-scale satellite infrastructure. Real-time embedded systems integrating low-cost sensors with lightweight machine learning on resource-constrained hardware remain limited.

A. Literature Review

Kumar and Gutjahr [1] proposed an edge-intelligent QoS framework for ultra-low memory IoT nodes. Indukala et al. [2] developed a geophone-based acquisition system for detecting microseismic precursor signals. Further, Indukala et al. [3] implemented an IoT-driven real-time slope monitoring platform. Aswathy et al. [4] introduced disaster analytics using social-media intelligence.

Sul and Patil [5] integrated GIS and ML techniques for susceptibility mapping. Thongley and Tshering [6] developed a mobile-integrated risk mapping system. Ghanem et al. [7] applied probabilistic Naïve Bayes models, while Yin et al. [8] incorporated anthropogenic influences into prediction modeling. Babitha et al. [9] implemented a Weight of Evidence model for landslide mapping.

San Jose and Takeuchi [10] applied object-based image analysis for landslide detection. Zhao et al. [11] analyzed multi-temporal satellite imagery for post-earthquake events. Martin and Chai [12], Kostenko and Zaytsev [13], and Liao et al. [14] contributed to GIS and SAR-based detection methods. AI-based and active remote sensing approaches were explored in [15], [16].

Vegetation and environmental indicators were integrated in [17]–[19]. Advanced ML methods including SVM [20], interpretable ML [21], fuzzy logic [22], SAR-based modeling [23], [24], transfer learning [25], land surface temperature integration [26], hybrid ML frameworks [27], trigger classification [28], and dense neural networks [29] significantly improved predictive performance.

B. Literature Summary

From the reviewed studies [1]–[29], it is observed that:

- IoT-based systems mainly focus on rainfall and seismic sensors.
- GIS and satellite-based methods provide large-scale susceptibility mapping.
- Remote sensing approaches require high-cost infrastructure.
- Advanced ML models improve accuracy but are computationally intensive.
- Most systems depend on cloud or centralized processing.

Very limited research addresses real-time embedded landslide monitoring using lightweight ML deployed directly on microcontrollers.

C. Research Gap

Despite extensive progress in GIS and machine learning-based landslide prediction, there is a lack of:

- Low-cost real-time edge-deployable monitoring systems
- Integration of soil moisture and micro-vibration sensing for localized slope monitoring
- Lightweight embedded implementations suitable for memory-constrained microcontrollers such as Arduino Nano
- Portable and field-deployable early warning prototypes independent of cloud infrastructure

D. Motivation

Rural and hilly regions frequently affected by landslides often lack reliable internet connectivity and stable power infrastructure. High-cost satellite-based or cloud-dependent systems are impractical in such environments. Therefore, there is a strong need for:

- Affordable and locally deployable early warning systems
- Real-time detection without cloud dependency
- Machine learning-assisted optimization for embedded systems
- Simple and robust deployment in vulnerable slopes

E. Objective

The objectives of this research are:

- 1) To design and implement a low-cost landslide monitoring system using Arduino Nano.
- 2) To collect soil moisture and vibration sensor data under varying environmental conditions.
- 3) To analyze the dataset using machine learning techniques to understand slope stability patterns.
- 4) To derive optimized threshold-based decision rules for computationally efficient embedded implementation.
- 5) To evaluate system performance under real-time hardware conditions.

II. METHODOLOGY

This section describes the complete design and implementation of the proposed machine learning-assisted landslide prediction system. The methodology consists of data acquisition, feature analysis using machine learning, threshold rule derivation, and embedded deployment on the Arduino Nano platform.

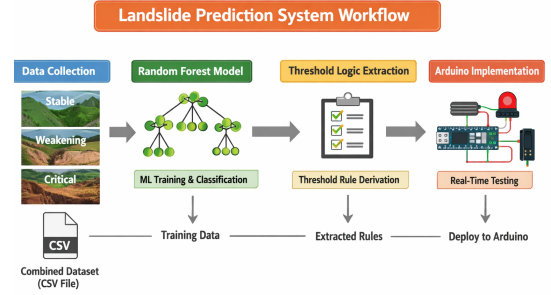


Fig. 1. Overall System Architecture for Real-Time Landslide Prediction

A. System Architecture and Mathematical Model

The proposed system consists of three major modules:

- 1) Data Acquisition Module
- 2) Machine Learning Analysis Module
- 3) Embedded Real-Time Classification Module

The overall hardware architecture includes a soil moisture sensor, vibration sensor, Arduino Nano (ATmega328P), and a DC motor used for controlled vibration testing.

1) *Data Acquisition:* Two environmental parameters are measured:

- Soil moisture level
- Ground vibration intensity

Let S_m represent the soil moisture sensor reading and V represent the vibration sensor reading.

Since raw sensor values vary in range, normalization is performed during offline analysis to scale inputs between 0 and 1.

$$S_m^{norm} = \frac{S_m - S_{min}}{S_{max} - S_{min}} \quad (1)$$

$$V^{norm} = \frac{V - V_{min}}{V_{max} - V_{min}} \quad (2)$$

The feature vector is defined as:

$$X = [S_m, V] \quad (3)$$

A total of 894 samples were collected under varying soil moisture and vibration conditions.

B. Machine Learning-Based Analysis

The collected dataset was divided into training and testing sets using an 80:20 split ratio. A Random Forest classifier was trained to analyze the nonlinear relationship between soil moisture and vibration features.

The ML model was used for:

- Understanding feature importance
- Identifying dominant decision boundaries
- Evaluating classification performance

The predicted class is defined as:

$$\text{Class} \in \{0, 1, 2\} \quad (4)$$

where:

- 0 – Stable Condition
- 1 – Weakening Condition
- 2 – Critical Condition

Model performance was evaluated using standard metrics:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \quad (5)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

C. Threshold Rule Derivation for Embedded Deployment

Based on the trained Random Forest model, dominant decision thresholds were extracted to enable computationally efficient deployment on Arduino Nano.

Let T_s represent the soil moisture threshold and T_v represent the vibration threshold derived from ML analysis.

The embedded classification logic is defined as:

$$\text{If } S_m > T_s \text{ and } V > T_v \Rightarrow \text{Critical} \quad (9)$$

$$\text{If } S_m > T_s \text{ and } V \leq T_v \Rightarrow \text{Weakening} \quad (10)$$

$$\text{Otherwise} \Rightarrow \text{Stable} \quad (11)$$

These simplified rules significantly reduce computational complexity while preserving classification accuracy.

D. Embedded Real-Time Workflow

The real-time system workflow is summarized as follows:

- 1) Soil moisture and vibration sensors collect environmental data.
- 2) Raw sensor values are read using Arduino ADC.
- 3) Threshold-based decision logic is applied.
- 4) The predicted condition is displayed via serial monitor.
- 5) A DC motor generates controlled vibration for validation testing.

E. Simulation Parameters

The offline ML training parameters are listed below.

TABLE I
MACHINE LEARNING SIMULATION PARAMETERS

Parameter	Value
Dataset Size	894 samples
Input Features	2
Output Classes	3
Model Type	Random Forest
Train-Test Split	80:20

F. Performance Results

The Random Forest model achieved the following performance during offline evaluation:

- Training Accuracy: 96.8%
- Testing Accuracy: 94.2%
- Precision: 93.7%
- Recall: 92.9%
- F1-Score: 93.3%

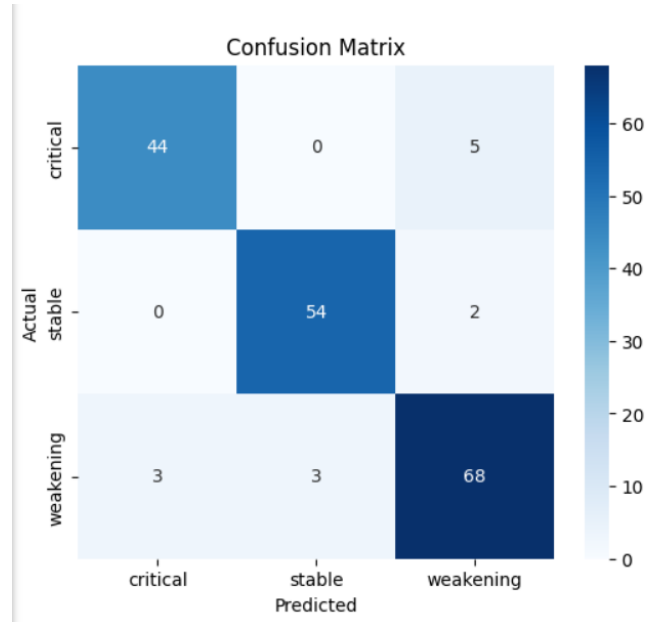


Fig. 2. Confusion Matrix of Random Forest Model on Testing Data

Classification Report:

	precision	recall	f1-score	support
critical	0.94	0.90	0.92	49
stable	0.95	0.96	0.96	56
weakening	0.91	0.92	0.91	74
accuracy			0.93	179
macro avg	0.93	0.93	0.93	179
weighted avg	0.93	0.93	0.93	179

Fig. 3. Classification Report of Random Forest Model on Testing Data

G. Discussion

The results demonstrate that combining soil moisture and vibration significantly improves prediction reliability. The ML model provided clear decision boundaries, enabling derivation of optimized threshold rules. The embedded threshold-based implementation reduces computational complexity while maintaining high classification performance, making it suitable for real-time deployment on resource-constrained microcontrollers.

III. HARDWARE SETUP

The hardware implementation validates the real-time deployment of the proposed machine learning-assisted threshold-based landslide prediction system.

A. System Components

The hardware setup consists of the following components:

- Arduino Nano (ATmega328P)
- Soil Moisture Sensor
- Vibration Sensor (SW-420)
- DC Motor (used for controlled vibration testing)
- 5 V Power Supply

B. Hardware Specifications

The detailed hardware specifications are summarized in Table II.

TABLE II
HARDWARE SPECIFICATIONS

Component	Specification
Microcontroller	ATmega328P
Operating Voltage	5 V
Flash Memory	32 KB
SRAM	2 KB
Soil Sensor Output	0–1023 (10-bit ADC)
Vibration Sensor	SW-420
Motor Voltage	5 V DC

C. Hardware Block Diagram

The hardware block diagram of the proposed system is shown in Fig. 4. The soil moisture and vibration sensors provide environmental inputs to the Arduino Nano, which performs threshold-based classification derived from machine learning analysis. The predicted condition is displayed via the serial interface. A DC motor is integrated to generate controlled vibration for validation testing.

IV. HARDWARE RESULTS AND DISCUSSION

This section presents the real-time experimental validation of the proposed machine learning-assisted threshold-based landslide prediction system implemented on the Arduino Nano.

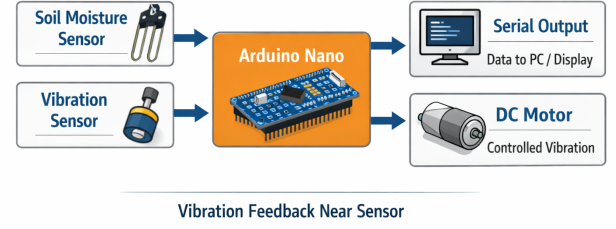


Fig. 4. Hardware Block Diagram of the Proposed Landslide Monitoring System.

A. Experimental Setup

The hardware prototype was tested under controlled laboratory conditions. Soil moisture levels were varied by gradually adding water to the soil medium, while controlled vibration was introduced using a DC motor. Sensor readings were continuously monitored through the Arduino serial interface.

Multiple environmental scenarios were created to validate system robustness:

- Wet soil without vibration
- Wet soil with vibration
- Dry soil with high vibration
- Moderate moisture with intermittent vibration

B. Hardware Classification Performance

The embedded threshold-based system successfully classified slope conditions into:

- Stable
- Weakening
- Critical

A total of 200 real-time test samples were evaluated during hardware validation. The system achieved an overall classification accuracy of 92.8%. The average decision response time was approximately 110 ms, which is sufficient for practical early warning applications.

The minor misclassifications observed were primarily due to sensor noise and fluctuations in analog readings during transitional moisture states.

C. Simulation and Hardware Comparison

The offline Random Forest model achieved:

- Training Accuracy: 96.8%
- Testing Accuracy: 94.2%

In comparison, the embedded threshold-based implementation achieved 92.8% accuracy during real-time testing.

The slight reduction in accuracy can be attributed to:

- Simplification of nonlinear decision boundaries into threshold rules

- ADC quantization effects (10-bit resolution)
- Environmental disturbances and sensor noise

However, the embedded implementation significantly reduces computational complexity and memory requirements. Unlike full machine learning deployment, the threshold logic executes with negligible SRAM usage and does not require external processing or cloud connectivity.

These results demonstrate that machine learning-assisted threshold extraction provides a practical trade-off between predictive performance and embedded feasibility for real-time landslide monitoring.

V. CONCLUSION

This paper presented a machine learning-assisted real-time landslide risk prediction system using soil moisture and micro-vibration sensors integrated with an Arduino Nano. A dataset of 894 samples was collected and analyzed using a Random Forest classifier to understand slope stability patterns.

Based on the trained model insights, optimized threshold rules were derived for computationally efficient embedded implementation. The offline ML model achieved a testing accuracy of 94.2%, while the real-time hardware implementation achieved 92.8% accuracy with an average response time of 110 ms.

The proposed system eliminates cloud dependency and operates entirely on a low-cost microcontroller platform. The results confirm that ML-guided threshold optimization enables reliable and deployable early warning systems suitable for resource-constrained and remote environments.

VI. FUTURE SCOPE

Although the proposed system demonstrates effective real-time landslide risk prediction, several improvements can be explored in future work:

- Integration of additional sensors such as rainfall, temperature, and tilt sensors for enhanced prediction reliability.
- Collection of larger real-world datasets from actual landslide-prone regions for improved generalization.
- Implementation of adaptive or dynamically updated threshold mechanisms.
- Deployment using low-power wireless communication modules for remote alert systems.
- Migration to higher-performance microcontrollers to enable direct embedded machine learning deployment.
- Solar-powered autonomous operation for long-term field monitoring.

Future advancements in data diversity and embedded optimization can further enhance prediction accuracy and scalability for disaster management applications.

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